

Task 10 — AI-Assisted Code Review, Debugging & Test Generation

Sub-Task A — Code Review Prompt

Act as a Python developer. Review the `calculate_discount` function for correctness, edge cases, security and PEP8. Explain the problems simply and show improved code.

What did AI find?

AI found the discount calculation bug. It also suggested validating cart items, checking null values, and keeping the code clear and PEP8-friendly.

Sub-Task B — Bug Investigation Prompt

Find the logic bug in this function. Explain why the calculation is wrong and give me a fixed version. Keep the explanation simple.

Bug Result

The code subtracts the discount rate directly. A value like 0.10 means 10%, not 0.10 currency units. Correct idea: `total = total - (total * discount)`.

Fixed Version

```
def calculate_discount(cart_items, promo_code):
    promo_codes = {'SAVE10': 0.10, 'SAVE20': 0.20, 'VIP50': 0.50}
    total = sum(item['price'] * item['quantity'] for item in cart_items)
    if promo_code in promo_codes:
        total -= total * promo_codes[promo_code]
    return max(total, 0)
```

Sub-Task C — Test Generation Prompt

Generate simple pytest unit tests for the fixed `calculate_discount` function. Cover happy path, empty input, null values and boundary conditions.

What did AI produce?

Tests for a normal cart with promo code, empty cart, invalid/null values, zero quantities and different discount codes.

Reflection

Bug investigation surprised me most because AI found the calculation problem quickly and explained why it was wrong.